Criteria	Weightage	Marks
Requirements Analysis & Conceptual Understanding	20%(6)	
System Design & Simulation Model Development	25% (7.5)	
Implementation & Tool Usage	20% (6)	
Testing, Validation & Result Analysis	20% (6)	
Documentation	15%(4.5)	

Code of Conduct:

I ___Rakshitha S / 192411008__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rakshitha S

1.AIM :

To design and simulate an Embedded C-based Vibration Monitoring Node using the AT89C51 microcontroller in Keil µVision, integrating periodic sensor sampling, timer-interrupt-driven scheduling, real-time vibration data processing, threshold-based status detection, and UART-based data logging.

2. PSEUDO CODE :

```
START
Initialize AT89C51 ports
Initialize UART communication
Set baud rate to 9600 bps
Initialize Timer 0
Enable Timer 0 interrupt
Enable global interrupts
Display system initialization message
REPEAT FOREVER
  IF sampling flag is set THEN
    Clear sampling flag
    Generate/read simulated vibration sensor value
    IF vibration level is NORMAL THEN
      Turn OFF warning LED
      Turn OFF alarm LED
      Turn OFF buzzer
      Display NORMAL status through UART
    ELSE IF vibration level is HIGH THEN
      Turn ON warning LED
      Turn ON alarm LED
      Turn ON buzzer
      Display WARNING status through UART
      Display excessive vibration alert
    END IF
```

```
    END IF

END REPEAT
Timer Interrupt:
    Reload Timer 0
    Increment timer counter
    IF timer counter reaches sampling interval THEN
        Reset timer counter
        Set sampling flag
    END IF
STOP
```

3. EMBEDDED C CODE :

```c
#include <reg51.h>

/*=================================================
  VIBRATION MONITORING NODE
  Controller : AT89C51
  Simulation : Keil µVision
  UART       : 9600 Baud
=================================================*/

/* Output pins */
sbit WARNING_LED = P2^0;
sbit ALARM_LED   = P2^1;
sbit BUZZER      = P2^2;

/* Global Variables */
unsigned int timer_count = 0;
unsigned char simulation_count = 0;
```

```
bit sample_flag = 0;

/*======================================================
  UART INITIALIZATION
======================================================*/
void UART_Init(void)
{
    TMOD |= 0x20;       /* Timer 1, Mode 2 */
    TH1 = 0xFD;         /* 9600 baud @ 11.0592 MHz */
    SCON = 0x50;        /* UART Mode 1 */
    TR1 = 1;            /* Start Timer 1 */

    TI = 1;
}

/*======================================================
  UART TRANSMIT CHARACTER
======================================================*/
void UART_TxChar(unsigned char ch)
{
    SBUF = ch;

    while(TI == 0);

    TI = 0;
}

/*======================================================
  UART TRANSMIT STRING
```

```
=====================================================*/
void UART_TxString(char *str)
{
    while(*str)
    {
        UART_TxChar(*str);
        str++;
    }
}

/*=====================================================
  TIMER 0 INITIALIZATION
=====================================================*/
void Timer0_Init(void)
{
    TMOD &= 0xF0;
    TMOD |= 0x01;       /* Timer 0 Mode 1 */

    /* Approximately 1 ms interrupt */
    TH0 = 0xFC;
    TL0 = 0x66;

    ET0 = 1;            /* Enable Timer 0 interrupt */
    EA  = 1;            /* Enable global interrupts */

    TR0 = 1;            /* Start Timer 0 */
}

/*=====================================================
  TIMER 0 INTERRUPT SERVICE ROUTINE
```

```
=======================================================*/
void Timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC;
    TL0 = 0x66;

    timer_count++;

    /* Approximately 1 second sampling interval */
    if(timer_count >= 1000)
    {
        timer_count = 0;
        sample_flag = 1;
    }
}

/*=======================================================
  SIMULATED VIBRATION SENSOR
=======================================================*/
unsigned char Read_Vibration(void)
{
    simulation_count++;

    /* Samples 1-3: Normal vibration */
    if(simulation_count <= 3)
    {
        return 0;
    }

    /* Samples 4-6: High vibration */
```

```
    else if(simulation_count <= 6)
    {
        return 1;
    }

    /* Return to normal */
    else
    {
        simulation_count = 0;
        return 0;
    }
}

/*==================================================
  PROCESS VIBRATION DATA
==================================================*/
void Process_Vibration(void)
{
    unsigned char vibration;

    vibration = Read_Vibration();

    UART_TxString("\r\n");
    UART_TxString("--------------------------------\r\n");

    if(vibration == 0)
    {
        WARNING_LED = 0;
        ALARM_LED   = 0;
        BUZZER      = 0;
```

```
        UART_TxString("VIBRATION LEVEL : NORMAL\r\n");
        UART_TxString("STATUS          : SAFE\r\n");
    }
    else
    {
        WARNING_LED = 1;
        ALARM_LED   = 1;
        BUZZER      = 1;

        UART_TxString("VIBRATION LEVEL : HIGH\r\n");
        UART_TxString("STATUS          : WARNING\r\n");
        UART_TxString("ALERT           : EXCESSIVE VIBRATION\r\n");
    }

    UART_TxString("--------------------------------\r\n");
}

/*=====================================================
  SYSTEM INITIALIZATION
=====================================================*/
void System_Init(void)
{
    P2 = 0x00;

    UART_Init();
    Timer0_Init();

    UART_TxString("\r\n");
    UART_TxString("================================\r\n");
```

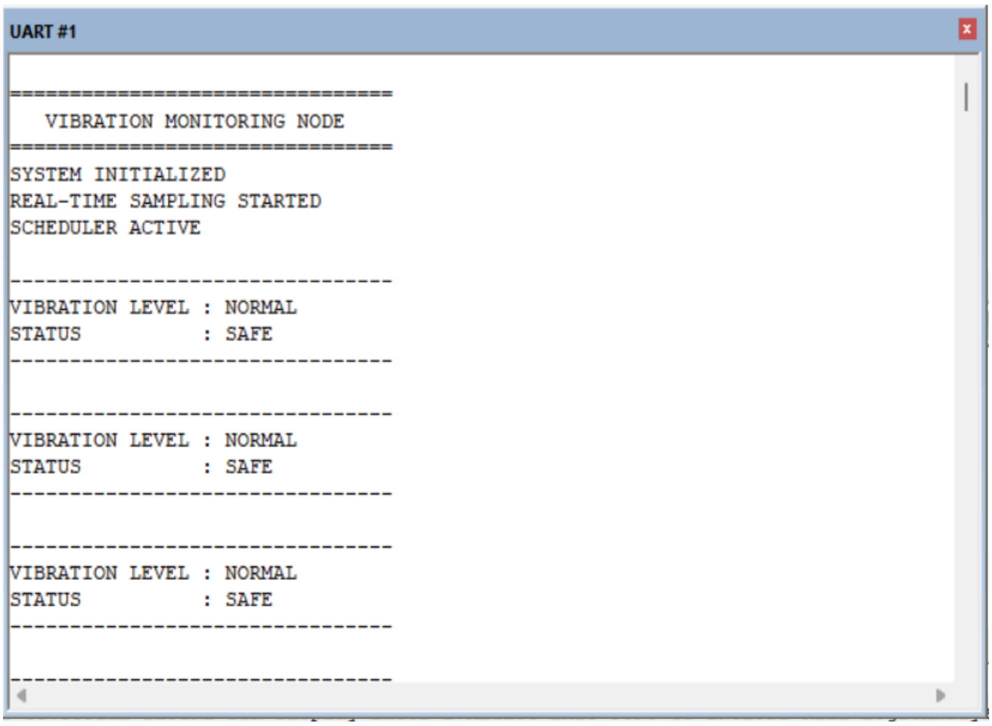
```
    UART_TxString("    VIBRATION MONITORING NODE\r\n");
    UART_TxString("===================================\r\n");
    UART_TxString("SYSTEM INITIALIZED\r\n");
    UART_TxString("REAL-TIME SAMPLING STARTED\r\n");
    UART_TxString("SCHEDULER ACTIVE\r\n");
}

/*======================================================
  MAIN PROGRAM
======================================================*/
void main(void)
{
    System_Init();

    while(1)
    {
        /* Scheduler */
        if(sample_flag == 1)
        {
            sample_flag = 0;

            Process_Vibration();
        }
    }
}
```

4 . OUTPUT :



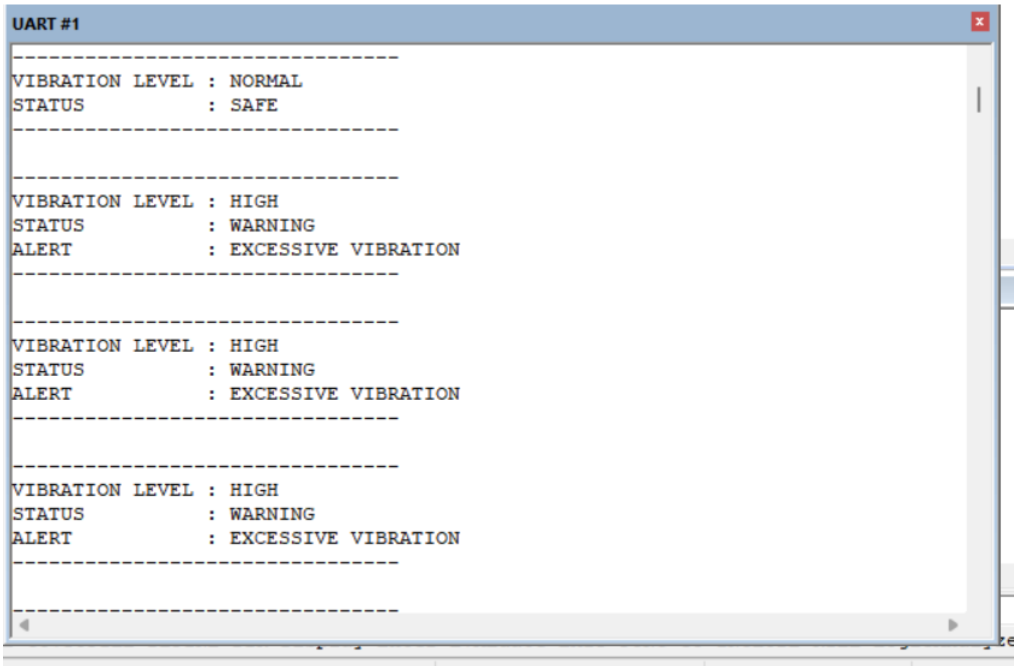
```
UART #1
================================
   VIBRATION MONITORING NODE
================================
SYSTEM INITIALIZED
REAL-TIME SAMPLING STARTED
SCHEDULER ACTIVE

--------------------------------
VIBRATION LEVEL : NORMAL
STATUS          : SAFE
--------------------------------

--------------------------------
VIBRATION LEVEL : NORMAL
STATUS          : SAFE
--------------------------------

--------------------------------
VIBRATION LEVEL : NORMAL
STATUS          : SAFE
--------------------------------

--------------------------------
```



```
UART #1
--------------------------------
VIBRATION LEVEL : NORMAL
STATUS          : SAFE
--------------------------------

--------------------------------
VIBRATION LEVEL : HIGH
STATUS          : WARNING
ALERT           : EXCESSIVE VIBRATION
--------------------------------

--------------------------------
VIBRATION LEVEL : HIGH
STATUS          : WARNING
ALERT           : EXCESSIVE VIBRATION
--------------------------------

--------------------------------
VIBRATION LEVEL : HIGH
STATUS          : WARNING
ALERT           : EXCESSIVE VIBRATION
--------------------------------

--------------------------------
```

5. RESULT :

The Vibration Monitoring Node was successfully designed and simulated in Keil µVision using Embedded C. Timer-interrupt-based scheduling and real-time vibration monitoring were implemented, and the sensor status was successfully displayed through UART #1.